

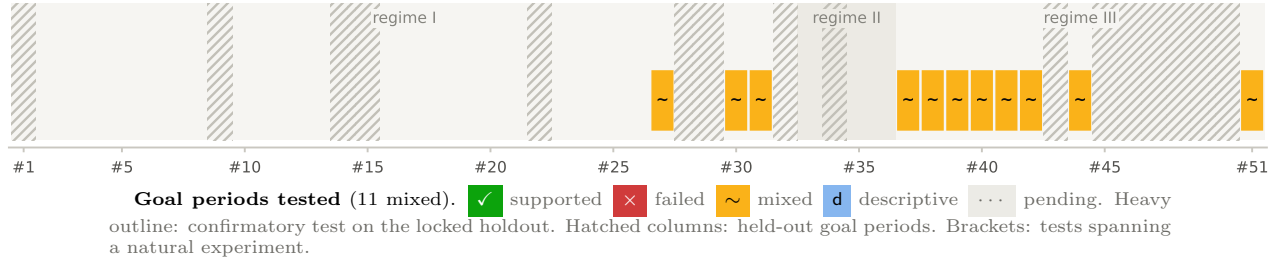
H16: Agents get stuck in metastable traps and escape by Kramers kinetics; nudges lower the barrier

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

Complete 42% Faithfulness 1.5/5 Usefulness 2.5/5

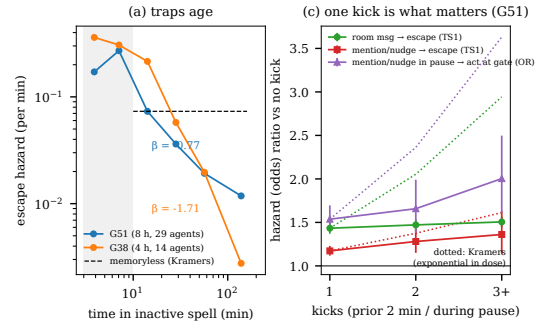
Bottom line. Agent traps are real and individual, but they age instead of escaping by Kramers kinetics, and kicks act as saturating triggers at scheduled decision points rather than as barrier-lowering forces.

level: hypothesis · pipeline checks 86% · updated 2026-10-03



WHAT HAPPENED

Four trap types (inactive spells, timer-pause chains, error loops, repeated-command loops) were tested in 11 non-holdout goal periods (3 regime I, 8 regime III). Predictions were written after a synthetic validation and before touching data. Kramers predicts memoryless escape; instead, traps *age*. Within agent, the escape hazard falls roughly as $t^{-0.8}$ (G51: $\beta = -0.77 [-0.81, -0.73]$, 4,321 escapes; G38: -1.71). Each re-pause lowers the odds of acting at the next timer expiry (G51: -0.38), and error loops age too (-0.72). Kicks help but saturate: one directed message during a pause raises the odds of acting $\times 1.5$ – 2.9 (5/7 periods), and more messages add little (14/16 powered dose curves additive or saturating, not exponential). Messages never broke error loops (0/10) and did not shorten long regime-I silences (0/3). $\beta J_0 = 0.09$ – 0.41 ; swarm activity is unimodal. The barrier was unidentifiable. Holdout (#32, #45): not run.



Observables. (a) Escape hazard from inactive spells vs. time in the spell (robust spells, ignoring single stray actions) in the two best-powered periods. Kramers predicts the flat dashed line once the first minutes (grey) have passed; instead the hazard falls, so traps age. (b) G51 hazard and odds ratios by number of kicks. Dotted lines extrapolate Kramers (exponential in dose) from one kick. The observed curves flatten: one message is what matters.

PHYSICAL MODEL

Models 01/02 (mean field) plus Kramers escape. Agent activity x moves in a double well $U(x)$ (idle and active basins) at temperature T . A trap is a stay in the idle basin, escaped at rate k over barrier ΔU (attempt rate ν). A kick (a message reaching the agent) adds force f . Swarm: Curie–Weiss coupling J_0 , field h , active fraction m . Fitted: hazard h_{it} with agent effects α_i , aging slope β , dose effects γ_n .

$$k = \nu e^{-\Delta U/T}, \quad G(x) = -\ln P(x) = U(x)/T + \text{const}$$

$$k(n) = k_0 e^{n f/T} \text{ (Kramers)} \quad \text{vs} \quad k(n) = k_0(1 + c n) \text{ (triggers)}$$

$$\ln[-\ln(1 - h_{it})] = \alpha_i + \beta \ln t + \sum_n \gamma_n \mathbb{K}[\text{dose}_{it} = n]$$

$$f(m) = -\frac{1}{2} J_0 m^2 - h m - T s(m); \text{ bistable iff } \beta J_0(1 - m^2) > 1$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Modestly useful, mostly as a correction. Facts an operator could use: a stuck agent gets stickier the longer it sits, so intervene early rather than wait. The lever that works is one directed message landing during a pause; the agent reads it at its next timer expiry. Repeating it, or chatting at the room, adds little. Chat does nothing for error loops, which need another fix (tooling, a reset). There is no swarm-wide tipping point at today's coupling. The physics added little: “Kramers” failed, the free-energy barrier was a smoothing artifact, and “aging plus saturating triggers” is ordinary survival analysis. Effects are modest and unconfirmed on the holdout.

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RESULTS VS PREDICTIONS

Pending.

GOAL PERIODS IN DETAIL

G27 ~ mixed
G30 ~ mixed
G31 ~ mixed
G37 ~ mixed
G38 ~ mixed
G39 ~ mixed
G40 ~ mixed
G41 ~ mixed
G42 ~ mixed
G44 ~ mixed
G51 ~ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	1
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Pending.

ROUND 2

- **Essence.** Traps deepen by self-reinforcement as the context fills with the agent’s own output.
- **H16-R1** Pólya urn: $P(\text{repeat}) \propto$ the share of the context that is own repeats.
- **H16-R2** One directed message works because it dilutes the urn; repeated messages add little once the context is diluted.
- **H16-R3** Trap depth is a model trait: some models loop more.

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